

ALLAN SEBASTIAN

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PROFESSIONAL SUMMARY

Computer Science engineer specialising in blockchain technology, decentralised systems, and full-stack web development. Published researcher in Ethereum-based social media architecture. Experienced in Solidity smart contracts, React frontend development, and security research focusing on open-source supply chain vulnerabilities. Seeking to apply decentralised systems expertise to products that give users verifiable control over their data.

TECHNICAL SKILLS

Web Development: HTML, CSS, JavaScript, ReactJS, Java, SQL

Systems & Languages: C, C++, Python, Java, SQL, Solidity

Blockchain & Web3: Solidity, Ethereum, Smart Contracts, Web3

Security & Research: Ethical Hacking, Malware Analysis, Vulnerability Research

Foundations: Data Structures & Algorithms, Operating Systems, Computer Networks, Machine Learning

PROJECT WORK

Decentralized Social Media Website

ReactJS, Solidity, Ethereum Blockchain

- Developed a blockchain-based social platform enabling users to post content and tip each other using Ethereum, eliminating centralised data intermediaries
- Authored Solidity smart contracts handling on-chain post storage and peer-to-peer tipping without reliance on a central authority
- Built the ReactJS frontend with wallet integration, allowing verifiable content ownership on the Ethereum network
- Project formed the basis of a peer-reviewed publication in IJIRCCE (May 2024)

Bad Snake — PyPI Malware Scanning Research

Python, Security Research, Supply Chain Analysis

- Investigated the vulnerability of open-source package repositories like PyPI to malware injection attacks
- Catalogued common attack vectors against open-source package indexes and analysed detection gaps
- Prototyped detection improvements for malicious Python packages and advocated stronger collaboration between security researchers and maintainers

Smart Medical Prescription Service

Web Platform, MySQL, Cloud-based Architecture

- Developed a cloud-based prescription management system bridging doctors and patients via a web interface
- Designed a MySQL database schema with unique patient codes for organised, retrievable prescription records
- Deployed as a cloud-based platform enabling anywhere-access for both clinicians and patients

RESEARCH PUBLICATION

Decentralized Social Media Using Ethereum Blockchain and Solidity Smart Contract

International Journal of Innovative Research in Computer and Communication Engineering (IJIRCCE) | Volume 12, Issue 5 | May 2024

Investigated how centralised social media companies harvest and monetise personal data, proposing a decentralised alternative built on Ethereum. The system uses Solidity smart contracts to handle content ownership and tipping without a central authority, giving users verifiable control over their data and interactions.

EDUCATION

B.Tech in Computer Science and Engineering

Prist College of Engineering | 2020 — 2024

Relevant Coursework: Object Oriented Programming, Databases, Data Structures & Algorithms, Operating Systems, Computer Networks, Machine Learning, Advanced DSA, Information Retrieval, Image Processing

Higher Secondary Education

Board of Higher Secondary Examination, Kerala | 2019 — 2020

SSLC (Secondary School Leaving Certificate)

Board of Public Examinations, Kerala | 2018